

Evaluating Missing Data Handling Techniques Under Different Missingness Mechanisms

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Introduction

Overview of Missingness Mechanisms

Missing data, whether due to corruption or logistical issues, can bias estimates, inflate standard errors, and invalidate inferences if not properly handled^[1]. Rubin's (1976) framework^[2] defines three mechanisms: Missing Completely at Random (MCAR), where missingness is unrelated to any data; Missing at Random (MAR), where missingness depends only on observed data; and Missing Not at Random (MNAR), where missingness depends on unobserved values, posing the greatest challenge and requiring strong, often unverifiable assumptions^[3]. Complete-case analysis is commonly used in applied research for its simplicity, but under MAR or MNAR conditions, it can introduce bias and reduce efficiency^[4]. Single imputation methods (e.g., mean substitution) also perform poorly, distorting distributions and underestimating uncertainty. In contrast, Multiple Imputation by Chained Equations (MICE) and related model-based approaches simulate missing values multiple times and pool results to reflect imputation uncertainty^[5]. While increasingly popular, the performance of these methods varies depending on the nature and extent of missingness mechanism^[6]. Simulation studies allow systematic evaluation of these methods by generating complete data, imposing missingness mechanisms, and comparing outcomes such as bias, confidence interval coverage, prediction accuracy, and model fit (e.g., AIC, BIC)^[7,8].

Relevance to Obesity Research

Obesity presents a growing public health challenge worldwide, contributing to increased risk of cardiovascular disease, type 2 diabetes, hypertension, musculoskeletal disorders, and certain cancers. According to the World Health Organization (2024)^[9], global obesity prevalence has tripled since 1975, and over 650 million adults are currently classified as obese. In Latin America, where countries like Mexico, Peru, and Colombia are experiencing rapid nutrition and lifestyle transitions, obesity rates have surged across all age groups^[10–12]. The determinants of obesity are multifactorial, including sociodemographic variables, poor dietary patterns, inadequate physical activity, sedentary behaviors, and harmful habits such as smoking and alcohol consumption^[13,14]. These risk factors are often collected via self-report or web-based tools, making them especially vulnerable to missingness due to social desirability bias, recall error, or issues with technology access. Understanding how missing data may bias the estimation of such relationships is critical for valid epidemiologic inference.

Study Objective

The objective of this project is to conduct a simulation study using the [UCI Obesity](#) dataset to assess how various imputation techniques perform under MCAR, MAR, and MNAR scenarios.

Methods

Data Source

We used the `obesity.levels` dataset from the `fairml` R package, which contains 2,111 records from individuals in Mexico, Peru, and Colombia. The dataset includes 17 demographic, behavioral, and physical condition attributes used to estimate obesity levels, labeled by the `NObeyesdad` variable. Approximately 77% of the data were synthetically generated using Weka and SMOTE, while 23% were collected via a web platform. Further details are elsewhere^[13,14]. The binary outcome variable, denoted as `Obesity`, was derived from the original multiclass label `NObeyesdad` by coding individuals with Obesity Type I, II, or III as “Obese” and others as “Not Obese”. Continuous predictors included: Age, FCVC (Frequency of vegetable consumption), NCP (Number of main meals), CH2O (daily water intake), FAF (physical activity frequency), and TUE (time using technology). Categorical variables included Gender, family history with overweight, high-calorie food consumption (FAVC), food between meals (CAEC), smoking (SMOKE), calorie monitoring (SCC), alcohol consumption (CALC), and transportation type (MTRANS).

Missing Data Generation Mechanisms

We simulated missingness under three mechanisms: To simulate MCAR, we randomly introduced missing values in three variables; `Gender`, `FAF`, and `CH2O`, with predefined rates of 15%, 25%, and 20%, respectively. The probability of missingness in these variables is independent of any observed or unobserved data. We simulated MAR by conditioning missingness on observed covariates. Specifically: `FAF` was more likely to be missing for older males; `SCC` was more likely to be missing among participants reporting high-calorie food consumption; `NCP` was more likely to be missing for younger females. Probabilities of missingness were generated using logistic functions of these predictors. For MNAR, missingness depended on unobserved (true) values of the variables themselves: `FAF` missingness was more likely in individuals with low physical activity; `CALC` was more likely missing in frequent drinkers; `SCC` missingness was higher among participants who did not monitor calories. These dependencies reflect real-world challenges where values themselves influence their probability of being missing.

Imputation Methods

We applied three common strategies to handle missing data; (1) Complete Case Analysis where all records with missing values in any of the selected predictors were excluded; (2) Mean/Mode Imputation where for continuous variables, missing values were imputed using the mean of observed values and for categorical variables, the most frequent category (mode) was used; (3) MICE using the `mice` package was used to perform 5 multiple imputations using chained equations.

Model Fitting and Evaluation

We trained logistic regression models to predict `Obesity` across all datasets (Original, MCAR, MAR, MNAR) and imputation methods. Each dataset was split into training (70%) and test (30%) sets using stratified sampling to preserve the outcome distribution. Model performance was evaluated using the following metrics: **Accuracy**: Proportion of correct classifications; **Sensitivity (Recall)**: Proportion of obese individuals correctly identified; **Specificity**: Proportion of non-obese individuals correctly identified; **Area Under the ROC Curve (AUC)**: Discriminative ability of the model; **F1 Score**: Harmonic mean of precision and recall. Model metrics were computed separately for the training and test sets. For MICE, pooled predictions across imputations were used for test metrics, and average metrics were reported for training.

Results

Exploratory Data Analysis

We conducted exploratory analysis to examine predictor distributions and associations by obesity status. The outcome was moderately balanced (54% Not Obese, 46% Obese). A correlation plot revealed notable differences by class; for example, Age and TUE were negatively correlated ($r = -0.297$ overall), more so among obese participants. Relationships between FCVC and NCP also varied across groups.

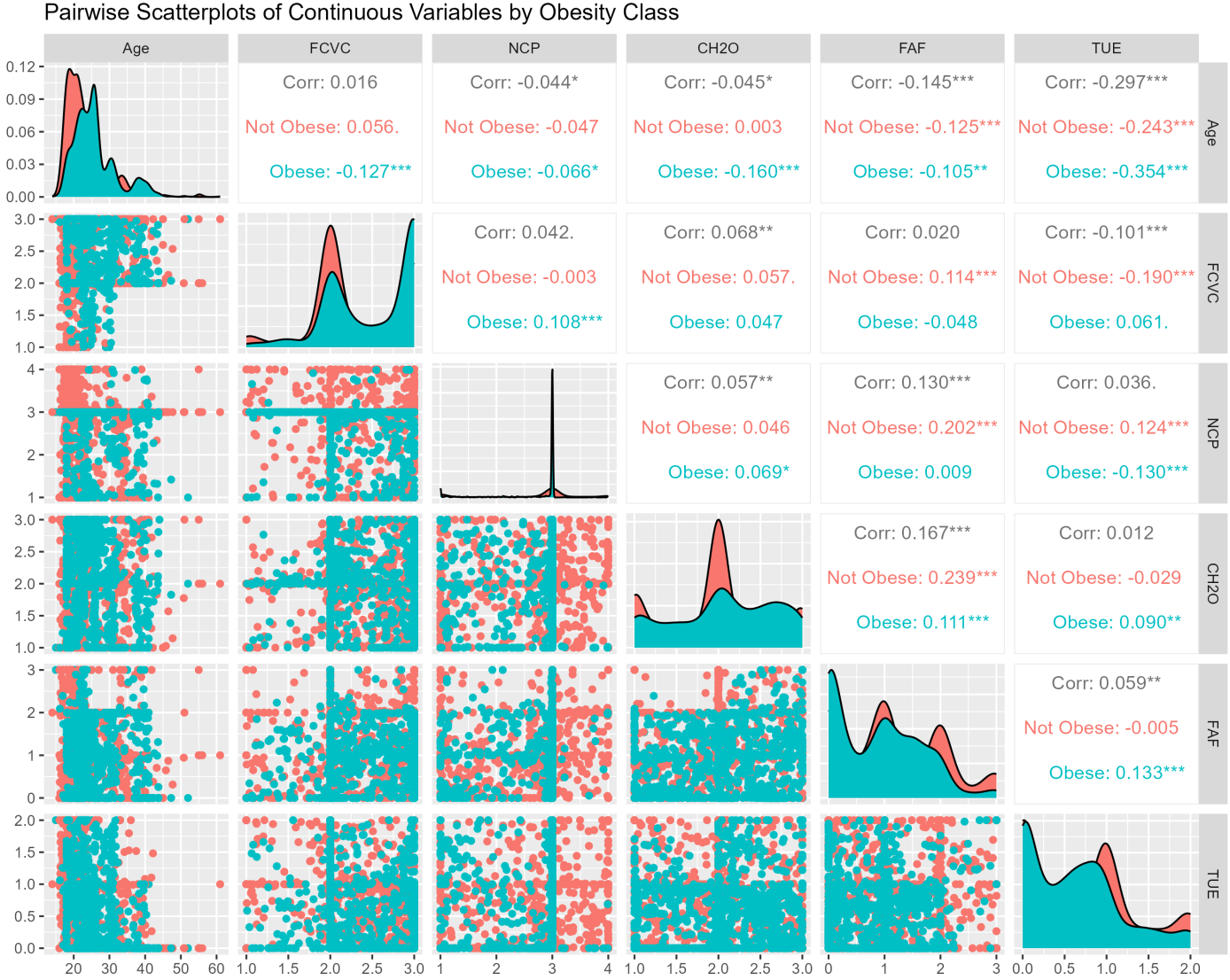


Figure 1: Exploratory Data Analysis of Continuous Predictors by Obesity Status

We used Little's MCAR test (`nanianar`, `misty`) to assess missingness in the simulated dataset. Results ($\chi^2 = 103.68$, $df = 93$, $p = 0.211$) suggest the data was consistent with MCAR. There were 8 unique missing patterns, with 1031 of 2111 cases missing at least one value. Formal tests for MAR or MNAR were not possible; distinguishing these mechanisms requires unverifiable assumptions or external data. Missingness patterns are shown in the figure 2 below.

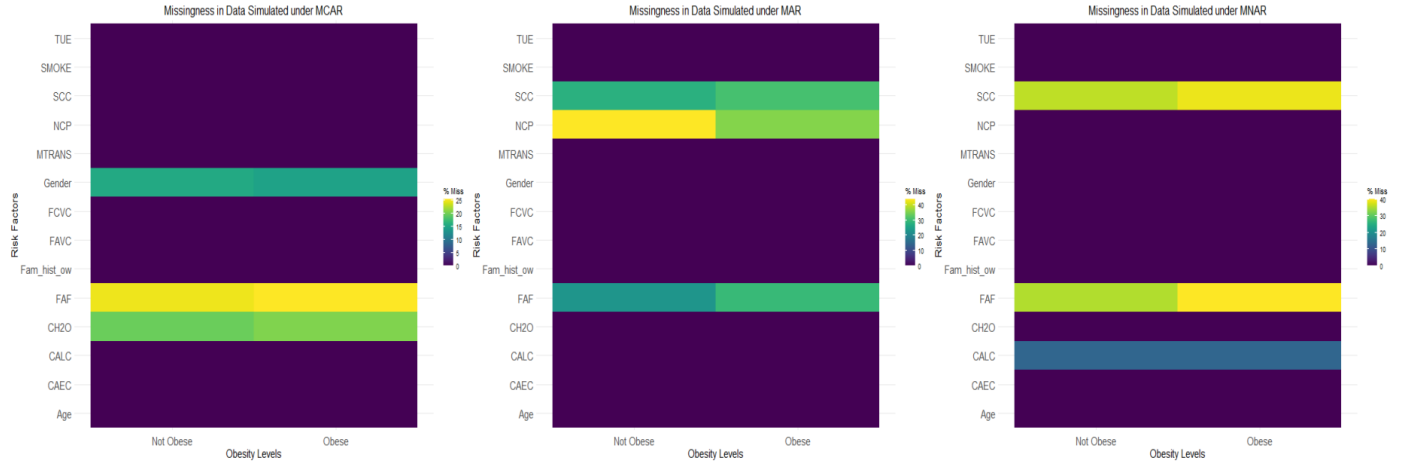


Figure 2: Missingness Patterns under MCAR, MAR, and MNAR

Training Set Performance

On the original full dataset without missing values, the model achieved an accuracy of 77.5%, sensitivity of 86.6%, specificity of 69.7%, and an AUC of 0.8683. Under MCAR conditions, model performance varied only slightly across imputation methods. Multiple imputation produced results comparable to the original dataset (accuracy = 77.5%, AUC = 0.8705), whereas complete case analysis and mean/mode imputation led to modest reductions in performance metrics. Notably, analysis with simulated data under MCAR yielded an accuracy of 76.3% and an AUC of 0.8642, see Table 1.

Table 1: Model Performance on Training Set Across Missing Data Handling Methods

Scenario	Method	Accuracy	Sensitivity	Specificity	AUC	F1 Score
Full Dataset						
	Original Dataset (No Missing)	0.7748	0.8664	0.6967	0.8683	0.7799
MCAR						
	Simulated (with missing)	0.7627	0.8612	0.6751	0.8642	0.7735
	Complete Case	0.7649	0.8613	0.6837	0.8590	0.7700
	Mean/Mode Imputation	0.7694	0.8576	0.6942	0.8691	0.7740
	Multiple Imputation (MICE)	0.7753	0.8567	0.7058	0.8705	0.7783
MAR						
	Simulated (with missing)	0.7813	0.8135	0.7561	0.8829	0.7659
	Complete Case	0.8077	0.8182	0.7991	0.8871	0.7927
	Mean/Mode Imputation	0.7769	0.8678	0.6992	0.8700	0.7817
	Multiple Imputation (MICE)	0.7849	0.8658	0.7158	0.8700	0.7875
MNAR						
	Simulated (with missing)	0.7550	0.8145	0.7082	0.8720	0.7453
	Complete Case	0.7827	0.8505	0.7314	0.8865	0.7712
	Mean/Mode Imputation	0.7647	0.8517	0.6905	0.8660	0.7692
	Multiple Imputation (MICE)	0.7751	0.8614	0.7015	0.8678	0.7791

In the MAR scenario, performance was more variable. Mean/mode imputation and multiple imputation yielded high sensitivity (0.8678 and 0.8658, respectively), but complete case analysis achieved the highest accuracy

(80.8%) and AUC (0.8871). This may be due to selective retention of more informative observations in the complete-case subset under the MAR mechanism. Multiple imputation again demonstrated strong overall performance with a balance of sensitivity and specificity. For the MNAR scenario, overall performance was lower compared to MCAR and MAR, as expected given the unidentifiability of the missing data mechanism without further assumptions. Multiple imputation consistently outperformed the other methods, closely matching the original dataset’s accuracy and AUC. Mean/Mode Imputation under MNAR led to the lowest AUC (0.8660) among the imputation methods and lowest specificity (0.6905), see Table 1 above.

Test Set Performance

Similar trends were observed in the test set evaluations. Under MCAR, the highest sensitivity (90.7%) was observed in the simulated dataset with missingness. Multiple imputation again yielded performance close to the original data (accuracy = 77.7%, AUC = 0.8736), while complete case analysis showed a slight increase in the AUC (0.8924) estimate, see Table 2 below. Under MAR, mean/mode imputation achieved the accuracy (77.7%) slightly above the original estimate and competitive AUC (0.8711), albeit at the cost of lower specificity. Multiple imputation led to a balanced trade-off across all metrics. In the MNAR setting, performance was generally lower. The simulated incomplete data yielded an AUC of 0.8649, but this dropped to 0.8404 under complete-case analysis. Nonetheless, MICE showed the most stable performance across all metrics and missingness mechanisms, including MNAR, with an F1 Score of 0.7861 and accuracy of 78.6%.

Table 2: Model Performance on Test Set Across Missing Data Handling Methods

Scenario	Method	Accuracy	Sensitivity	Specificity	AUC	F1 Score
Full Dataset						
	Original Dataset (No Missing)	0.7753	0.8385	0.7214	0.8717	0.7746
MCAR						
	Simulated (with missing)	0.7909	0.9071	0.7053	0.8862	0.7864
	Complete Case	0.7740	0.8844	0.6818	0.8924	0.7808
	Mean/Mode Imputation	0.7737	0.8488	0.7097	0.8717	0.7755
	Multiple Imputation (MICE)	0.7769	0.8557	0.7097	0.8736	0.7793
MAR						
	Simulated (with missing)	0.7987	0.8082	0.7901	0.8791	0.7919
	Complete Case	0.7853	0.8354	0.7449	0.8776	0.7765
	Mean/Mode Imputation	0.7769	0.8625	0.7038	0.8711	0.7807
	Multiple Imputation (MICE)	0.7864	0.8660	0.7185	0.8770	0.7887
MNAR						
	Simulated (with missing)	0.7476	0.8452	0.6803	0.8649	0.7320
	Complete Case	0.7109	0.7692	0.6667	0.8404	0.6965
	Mean/Mode Imputation	0.7816	0.8591	0.7155	0.8721	0.7837
	Multiple Imputation (MICE)	0.7864	0.8522	0.7302	0.8765	0.7861

Conclusion

Multiple imputation (MICE) showed the most consistent performance across all missingness mechanisms. Complete case analysis performed well under MAR but led to bias and data loss under MCAR and MNAR. Mean/mode imputation was simple but less effective. These findings highlight the need for principled approaches when missingness may not be MCAR.

Data Availabiliy

All codes and datasets for this project can be found at our [GitHub Repository](#).

References

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Appendix: All code for this report

```
knitr::opts_chunk$set(echo = FALSE,
                      message = FALSE,
                      warning = FALSE,
                      dpi = 300)

# =====
# Load Required Libraries
# =====

library(mice)      # Multiple imputation using chained equations (MICE)
library(caret)     # Data splitting, model training, and performance metrics
library(pROC)      # ROC curve analysis and AUC calculations
library(tidyverse) # Data manipulation and visualization
```

```

library(naniar)      # Missing data visualization and diagnostics
library(fairml)      # Accessing the obesity.levels dataset used in this study
library(misty)       # Additional missing data analysis tools and diagnostics
library(mvnmle)      # Maximum likelihood estimation with missing values
library(ggplot2)     # For creating publication-quality visualizations
library(gridExtra)   # For arranging multiple ggplot2 plots in a grid layout
library(magick)       # For reading, editing, and saving images in R (e.g., PNG, JPG);
  ↪ supports image processing workflows.
library(grid)        # Base R package for raster images on a grid layout.
library(kableExtra)  # For formatting tables
library(GGally)      # Pairs Plot

# =====
# Load and Prepare the Dataset
# =====
data(obesity.levels) # Load dataset from fairml package

# Create binary obesity classification
obesity.levels$Obesity <- ifelse(obesity.levels$NObeyesdad %in%
                                c("Obesity_Type_I", "Obesity_Type_II",
                                  ↪ "Obesity_Type_III"),
                                1, 0)
obesity.levels$Obesity <- factor(obesity.levels$Obesity,
                                levels = c(0, 1),
                                labels = c("Not Obese", "Obese"))
obesity.levels <- obesity.levels %>%
  rename(Fam_hist_ow = family_history_with_overweight)

# Define variable types
cont_vars <- c("Age", "FCVC", "NCP", "CH2O", "FAF", "TUE")
cat_vars <- c("Gender", "Fam_hist_ow", "FAVC", "CAEC",
              "SMOKE", "SCC", "CALC", "MTRANS")
target_var <- "Obesity"

# Final dataset without Height/Weight
obesity_full <- obesity.levels %>%
  select(all_of(c(cont_vars, cat_vars, target_var)))

# =====
# Exploratory Data Analysis
# =====

# Outcome distribution

table(obesity_full$Obesity) # Not Obese: 1139      Obese: 972

# Proportions
prop.table(table(obesity_full$Obesity)) # Not Obese: 0.5395547  Obese: 0.4604453

# Box plots for a few key continuous variables

```

```

obesity_full %>%
  pivot_longer(cols = all_of(cont_vars), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Obesity, y = Value, fill = Obesity)) +
  geom_boxplot() +
  facet_wrap(~Variable, scales = "free", ncol = 3) +
  labs(title = "Distribution of Continuous Predictors by Obesity Status") +
  theme_minimal()

# Proportions of categorical variables by class
cat_summary <- function(var) {
  obesity_full %>%
    count(!sym(var), Obesity) %>%
    group_by(!sym(var)) %>%
    mutate(Percent = n / sum(n) * 100)
}

cat_vars %>% map(cat_summary)

# Pairs plot
ggpairs(obesity_full,
        columns = cont_vars,
        aes(color = Obesity),
        title = "Pairwise Scatterplots of Continuous Variables by Obesity Class")
ggsave("Analysis/pairs_plot.png", width = 10, height = 8)

knitr::include_graphics("Analysis/pairs_plot.png")

# =====
# Missing Data Simulation Functions
# =====

# MCAR: Missing completely at random
simulate_mcar <- function(data, vars, rates) {
  set.seed(123)
  mcar_data <- data
  for (i in seq_along(vars)) {
    var <- vars[i]
    rate <- rates[i]
    missing_idx <- sample(nrow(data), size = floor(rate * nrow(data)))
    mcar_data[missing_idx, var] <- NA
  }
  return(mcar_data)
}

# MAR: Missing at random (depends on other observed variables)
simulate_mar <- function(data) {
  set.seed(123)
  mar_data <- data

  # FAF missingness: Higher in older males

```

```

age_scaled <- scale(data$Age)
male <- as.numeric(data$Gender == "Male")
logit_faf <- -1.5 + 0.8*age_scaled + 0.7*male
p_faf <- plogis(logit_faf)
missing_faf <- rbinom(nrow(data), 1, p_faf) == 1
mar_data$FAF[missing_faf] <- NA

# SCC missingness: Higher in those who consume high-calorie food
favc_yes <- as.numeric(data$FAVC == "yes")
logit_scc <- -2.0 + 1.2*favc_yes
p_scc <- plogis(logit_scc)
missing_scc <- rbinom(nrow(data), 1, p_scc) == 1
mar_data$SCC[missing_scc] <- NA

# NCP missingness: Higher in younger females
female <- as.numeric(data$Gender == "Female")
age_scaled <- scale(data$Age)
logit_ncp <- -1.0 - 0.9*age_scaled + 0.8*female
p_ncp <- plogis(logit_ncp)
missing_ncp <- rbinom(nrow(data), 1, p_ncp) == 1
mar_data$NCP[missing_ncp] <- NA

return(mar_data)
}

# MNAR: Missing not at random (depends on unobserved values)
simulate_mnar <- function(data) {
  set.seed(123)
  mnar_data <- data

  # FAF missingness: Higher in inactive people
  faf_low <- as.numeric(data$FAF < 1)
  logit_faf <- -1.5 + 1.8*faf_low
  p_faf <- plogis(logit_faf)
  missing_faf <- rbinom(nrow(data), 1, p_faf) == 1
  mnar_data$FAF[missing_faf] <- NA

  # CALC missingness: Higher in frequent drinkers
  calc_freq <- as.numeric(data$CALC == "Frequently")
  logit_calc <- -2.0 + 1.5*calc_freq
  p_calc <- plogis(logit_calc)
  missing_calc <- rbinom(nrow(data), 1, p_calc) == 1
  mnar_data$CALC[missing_calc] <- NA

  # SCC missingness: Higher in those who don't monitor calories
  scc_no <- as.numeric(data$SCC == "no")
  logit_scc <- -1.8 + 1.4*scc_no
  p_scc <- plogis(logit_scc)
  missing_scc <- rbinom(nrow(data), 1, p_scc) == 1
  mnar_data$SCC[missing_scc] <- NA
}

```

```

    return(mnar_data)
}

# Apply missingness mechanisms
mcar_data <- simulate_mcar(obesity_full,
                          vars = c("Gender", "FAF", "CH2O"),
                          rates = c(0.15, 0.25, 0.20))

mar_data <- simulate_mar(obesity_full)
mnar_data <- simulate_mnar(obesity_full)

# =====
# Diagnosing MCAR: Little's Test
# =====

# Perform Little's MCAR test using the `nanian` package
nanian::mcar_test(mcar_data)

# Perform the same test using the `misty` package for additional summary
misty::na.test(mcar_data)

# =====
# Visualize MCAR Missingness Patterns
# =====

# Heatmap of missing values
vis_miss(mcar_data) + ggtitle("MCAR Missingness Pattern")

# UpSet plot to show combinations of missingness
gg_miss_upset(mcar_data)

# Open a PNG device
png(filename = "Analysis/mcar_pattern.png", width = 800, height = 600)

# Visualize missingness across Obesity Levels
gg_miss_fct(x = mcar_data, fct = Obesity) +
  labs(title = "Missingness in Data Simulated under MCAR",
       x = "Obesity Levels", y = "Risk Factors") +
  theme(plot.title = element_text(hjust = 0.5, size = 16),
        axis.title.x = element_text(size = 15),
        axis.title.y = element_text(size = 15),
        axis.text.x = element_text(size = 15, angle = 0, vjust = 0.5, hjust = 0.5),
        axis.text.y = element_text(size = 15))

# Close the device (write file)
dev.off()

# =====
# Visualize MAR Missingness Patterns
# =====

```

```

# Heatmap of missing values
vis_miss(mar_data) + ggtitle("MAR Missingness Pattern")

# UpSet plot to show combinations of missingness
gg_miss_upset(mar_data)

# Open a PNG device
png(filename = "Analysis/mar_pattern.png", width = 800, height = 600)

# Visualize missingness across Obesity Levels
gg_miss_fct(x = mar_data, fct = Obesity) +
  labs(title = "Missingness in Data Simulated under MAR",
       x = "Obesity Levels", y = "Risk Factors") +
theme(plot.title = element_text(hjust = 0.5, size = 16),
      axis.title.x = element_text(size = 15),
      axis.title.y = element_text(size = 15),
      axis.text.x = element_text(size = 15, angle = 0, vjust = 0.5, hjust = 0.5),
      axis.text.y = element_text(size = 15))

# Close the device (write file)
dev.off()

# =====
# Visualize MNAR Missingness Patterns
# =====

# Heatmap of missing values
vis_miss(mnar_data) + ggtitle("MNAR Missingness Pattern")

# UpSet plot to show combinations of missingness
gg_miss_upset(mnar_data)

# Open a PNG device
png(filename = "Analysis/mnar_pattern.png", width = 800, height = 600)

# Visualize missingness across Obesity Levels
gg_miss_fct(x = mnar_data, fct = Obesity) +
  labs(title = "Missingness in Data Simulated under MNAR",
       x = "Obesity Levels", y = "Risk Factors") +
theme(plot.title = element_text(hjust = 0.5, size = 16),
      axis.title.x = element_text(size = 15),
      axis.title.y = element_text(size = 15),
      axis.text.x = element_text(size = 15, angle = 0, vjust = 0.5, hjust = 0.5),
      axis.text.y = element_text(size = 15))

# Close the device (write file)
dev.off()

knitr::include_graphics("Analysis/missing_patterns.png")

# =====

```

```

# Apply Imputation Methods
# =====

# Function to apply mean/mode imputation
mean_mode_impute <- function(data) {
  imp_data <- data
  # Impute continuous variables with mean
  for (var in cont_vars) {
    if (any(is.na(imp_data[[var]]))) {
      imp_data[is.na(imp_data[[var]]), var] <- mean(imp_data[[var]], na.rm = TRUE)
    }
  }
  # Impute categorical variables with mode
  for (var in cat_vars) {
    if (any(is.na(imp_data[[var]]))) {
      mode_val <- names(sort(table(imp_data[[var]]), decreasing = TRUE))[1]
      imp_data[is.na(imp_data[[var]]), var] <- mode_val
    }
  }
  return(imp_data)
}

# Apply imputation to all scenarios
impute_all <- function(scenario_data) {
  imputed <- list()

  # Complete case
  imputed$complete_case <- na.omit(scenario_data)

  # Mean/mode imputation
  imputed$mean_mode <- mean_mode_impute(scenario_data)

  # MICE imputation (5 imputations)
  mice_imp <- mice(scenario_data, m = 5, maxit = 10, printFlag = FALSE)
  imputed$mice <- lapply(1:5, function(i) complete(mice_imp, i))

  return(imputed)
}

# Apply to all scenarios
mcar_imputed <- impute_all(mcar_data)
mar_imputed <- impute_all(mar_data)
mnar_imputed <- impute_all(mnar_data)

# =====
# Data Splitting Function
# =====

split_dataset <- function(data, target = "Obesity", train_frac = 0.7) {
  set.seed(42)
  # Ensure both classes are present

```

```

idx <- createDataPartition(data[[target]], p = train_frac, list = FALSE)
train <- data[idx, ]
test  <- data[-idx, ]

# Check if both classes exist
if (length(unique(train[[target]])) < 2 | length(unique(test[[target]])) < 2) {
  # Re-sample if any set has only one class
  idx <- createDataPartition(data[[target]], p = train_frac, list = FALSE)
  train <- data[idx, ]
  test  <- data[-idx, ]
}

return(list(train = train, test = test))
}

# Split all datasets
splits <- list(
  full = split_dataset(obesity_full),

  mcar = list(
    original = split_dataset(mcar_data),
    complete_case = split_dataset(mcar_imputed$complete_case),
    mean_mode = split_dataset(mcar_imputed$mean_mode),
    mice = lapply(mcar_imputed$mice, split_dataset)
  ),

  mar = list(
    original = split_dataset(mar_data),
    complete_case = split_dataset(mar_imputed$complete_case),
    mean_mode = split_dataset(mar_imputed$mean_mode),
    mice = lapply(mar_imputed$mice, split_dataset)
  ),

  mnar = list(
    original = split_dataset(mnar_data),
    complete_case = split_dataset(mnar_imputed$complete_case),
    mean_mode = split_dataset(mnar_imputed$mean_mode),
    mice = lapply(mnar_imputed$mice, split_dataset)
  )
)

# =====
# Performance Evaluation Functions
# =====

# Corrected main performance function
run_logistic_performance <- function(train_data, test_data) {
  # Convert categorical to factors
  train_data <- train_data %>% mutate(across(all_of(cat_vars), as.factor))
  test_data <- test_data %>% mutate(across(all_of(cat_vars), as.factor))

  # Fit model

```

```

model <- glm(Obesity ~ ., data = train_data, family = binomial)

# Calculate metrics using corrected function
train_metrics <- calculate_metrics(model, train_data)
test_metrics <- calculate_metrics(model, test_data)

return(list(train = train_metrics, test = test_metrics))
}

# Metrics calculation function
calculate_metrics <- function(model, data) {
  prob_pred <- predict(model, newdata = data, type = "response")
  class_pred <- ifelse(prob_pred > 0.5, "Obese", "Not Obese")
  class_pred <- factor(class_pred, levels = c("Not Obese", "Obese"))

  # Handle cases where predictions don't cover all classes
  if (length(unique(class_pred)) < 2) {
    return(list(
      Accuracy = NA, Sensitivity = NA,
      Specificity = NA, AUC = NA, F1 = NA
    ))
  }

  cm <- confusionMatrix(class_pred, data$Obesity, positive = "Obese")
  auc_val <- auc(roc(data$Obesity, prob_pred))

  return(list(
    Accuracy = as.numeric(cm$overall["Accuracy"]),
    Sensitivity = as.numeric(cm$byClass["Sensitivity"]),
    Specificity = as.numeric(cm$byClass["Specificity"]),
    AUC = as.numeric(auc_val),
    F1 = as.numeric(cm$byClass["F1"])
  ))
}

# MICE-specific processing with Rubin's rules
process_mice_performance <- function(mice_splits) {
  models <- list()
  test_preds <- list()

  # Train models and collect test predictions
  for (i in 1:length(mice_splits)) {
    split <- mice_splits[[i]]

    # Convert categoricals to factors
    split$train <- split$train %>% mutate(across(all_of(cat_vars), as.factor))
    split$test <- split$test %>% mutate(across(all_of(cat_vars), as.factor))

    model <- glm(Obesity ~ ., data = split$train, family = binomial)
    models[[i]] <- model
  }
}

```

```

# Get test predictions
test_preds[[i]] <- predict(model, newdata = split$test, type = "response")
}

# Pool predictions using Rubin's rules
pooled_preds <- rowMeans(do.call(cbind, test_preds))

# Calculate metrics using pooled predictions
calculate_pooled_metrics <- function(preds, data) {
  class_pred <- ifelse(preds > 0.5, "Obese", "Not Obese")
  class_pred <- factor(class_pred, levels = c("Not Obese", "Obese"))

  if (nlevels(class_pred) < 2) {
    return(list(
      Accuracy = NA, Sensitivity = NA,
      Specificity = NA, AUC = NA, F1 = NA
    ))
  }

  cm <- confusionMatrix(class_pred, data$Obesity, positive = "Obese")
  auc_val <- auc(roc(data$Obesity, preds))

  return(list(
    Accuracy = as.numeric(cm$overall["Accuracy"]),
    Sensitivity = as.numeric(cm$byClass["Sensitivity"]),
    Specificity = as.numeric(cm$byClass["Specificity"]),
    AUC = as.numeric(auc_val),
    F1 = as.numeric(cm$byClass["F1"])
  ))
}

# Use first test set as reference
test_ref <- mice_splits[[1]]$test
test_metrics <- calculate_pooled_metrics(pooled_preds, test_ref)

# For training, average metrics across imputations
train_metrics_list <- map(models, ~ calculate_metrics(.x, .x$data))
train_metrics <- list(
  Accuracy = mean(map_dbl(train_metrics_list, "Accuracy"), na.rm = TRUE),
  Sensitivity = mean(map_dbl(train_metrics_list, "Sensitivity"), na.rm = TRUE),
  Specificity = mean(map_dbl(train_metrics_list, "Specificity"), na.rm = TRUE),
  AUC = mean(map_dbl(train_metrics_list, "AUC"), na.rm = TRUE),
  F1 = mean(map_dbl(train_metrics_list, "F1"), na.rm = TRUE)
)

return(list(train = train_metrics, test = test_metrics))
}

# =====
# Evaluate All Scenarios
# =====

```

```

results <- list()

# Full dataset baseline
results$full <- run_logistic_performance(splits$full$train, splits$full$test)

# Evaluate scenarios
scenarios <- c("mcar", "mar", "mnar")
methods <- c("complete_case", "mean_mode", "mice")

for (scenario in scenarios) {
  scenario_results <- list()

  # Original missing data
  scenario_results$original <- run_logistic_performance(
    splits[[scenario]]$original$train,
    splits[[scenario]]$original$test
  )

  # Imputation methods
  for (method in methods) {
    if (method == "mice") {
      # MICE requires special handling
      scenario_results[[method]] <- process_mice_performance(splits[[scenario]][[method]])
    } else {
      # Other methods
      split_data <- splits[[scenario]][[method]]
      scenario_results[[method]] <- run_logistic_performance(
        train_data = split_data$train,
        test_data = split_data$test
      )
    }
  }

  results[[scenario]] <- scenario_results
}

# =====
# Create Performance Tables
# =====
create_performance_table <- function(results, dataset = "test") {
  # Initialize data frame
  perf_df <- data.frame(
    Scenario = character(),
    Method = character(),
    Accuracy = numeric(),
    Sensitivity = numeric(),
    Specificity = numeric(),
    AUC = numeric(),
    F1 = numeric(),
    stringsAsFactors = FALSE
  )
}

```

```

# Add full dataset results
perf_df <- rbind(perf_df, data.frame(
  Scenario = "Full",
  Method = "Full Dataset",
  Accuracy = results$full[[dataset]]$Accuracy,
  Sensitivity = results$full[[dataset]]$Sensitivity,
  Specificity = results$full[[dataset]]$Specificity,
  AUC = results$full[[dataset]]$AUC,
  F1 = results$full[[dataset]]$F1
))

# Add scenario results
for (scenario in c("mcar", "mar", "mnar")) {
  # Original missing data
  perf_df <- rbind(perf_df, data.frame(
    Scenario = toupper(scenario),
    Method = "original",
    Accuracy = results[[scenario]]$original[[dataset]]$Accuracy,
    Sensitivity = results[[scenario]]$original[[dataset]]$Sensitivity,
    Specificity = results[[scenario]]$original[[dataset]]$Specificity,
    AUC = results[[scenario]]$original[[dataset]]$AUC,
    F1 = results[[scenario]]$original[[dataset]]$F1
  ))

  # Imputation methods
  for (method in methods) {
    perf_df <- rbind(perf_df, data.frame(
      Scenario = toupper(scenario),
      Method = method,
      Accuracy = results[[scenario]][[method]][[dataset]]$Accuracy,
      Sensitivity = results[[scenario]][[method]][[dataset]]$Sensitivity,
      Specificity = results[[scenario]][[method]][[dataset]]$Specificity,
      AUC = results[[scenario]][[method]][[dataset]]$AUC,
      F1 = results[[scenario]][[method]][[dataset]]$F1
    ))
  }
}

# Round values
perf_df[, 3:7] <- round(perf_df[, 3:7], 4)

return(perf_df)
}

# Create tables
training_table <- create_performance_table(results, "train")
test_table <- create_performance_table(results, "test")

# Format tables
training_table <- training_table %>%

```

```

rename(`F1 Score` = F1) %>%
select(Scenario, Method, Accuracy, Sensitivity, Specificity, AUC, `F1 Score`)

test_table <- test_table %>%
  rename(`F1 Score` = F1) %>%
  select(Scenario, Method, Accuracy, Sensitivity, Specificity, AUC, `F1 Score`)

training_table <- training_table %>%
  mutate(across(1, ~ "")) %>%
  mutate(Method = case_when(
    Method == "original" ~ "Simulated (with missing)",
    Method == "complete_case" ~ "Complete Case",
    Method == "mean_mode" ~ "Mean/Mode Imputation",
    Method == "mice" ~ "Multiple Imputation (MICE)",
    Method == "Full Dataset" ~ "Original Dataset (No Missing)",
    TRUE ~ Method))

# Displaying results
training_table %>%
  kbl(booktabs = TRUE, digits = 5, caption = "Model Performance on Training Set Across
  ↳ Missing Data Handling Methods") %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover", "condensed"),
  ↳ latex_options = "HOLD_position") %>%
  group_rows("Full Dataset", 1, 1) %>%
  group_rows("MCAR", 2, 5) %>%
  group_rows("MAR", 6, 9) %>%
  group_rows("MNAR", 10, 13)

test_table <- test_table %>%
  mutate(across(1, ~ "")) %>%
  mutate(Method = case_when(
    Method == "original" ~ "Simulated (with missing)",
    Method == "complete_case" ~ "Complete Case",
    Method == "mean_mode" ~ "Mean/Mode Imputation",
    Method == "mice" ~ "Multiple Imputation (MICE)",
    Method == "Full Dataset" ~ "Original Dataset (No Missing)",
    TRUE ~ Method))

# Displaying results
test_table %>%
  kbl(booktabs = TRUE, digits = 5,
    caption = "Model Performance on Test Set Across Missing Data Handling Methods") %>%
  kable_styling(full_width = FALSE,
    bootstrap_options = c("striped", "hover", "condensed"),
    latex_options = "HOLD_position") %>%
  group_rows("Full Dataset", 1, 1) %>%
  group_rows("MCAR", 2, 5) %>%
  group_rows("MAR", 6, 9) %>%
  group_rows("MNAR", 10, 13)

```